

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech (EC) Examination****December 2016****EC 407 Radar Systems****Date: 28.12.2016, Wednesday****Time: 01:30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below. [07]**

- a. In Direction Finders, the determination of direction is made by utilizing the directional property of _____ antenna or its modified form. [01]

- (a) Dipole antenna (b) Monopole antenna
(c) Loop antenna (d) Lens antenna

- b. Find the incorrect statement of the following regarding increasing loop voltage in Direction finders. [01]

- (a) Increase of Area of loop increase loop voltage (b) Increasing number of turns of antenna increases loop antenna
(c) Decreasing the Quality factor of loop increases Loop voltage (d) Increasing the Quality factor of loop increases Loop voltage

- c. In search Radar _____ antenna can be used. [01]

- (a) Wide beam (b) Wide and narrow beam both
(c) Narrow beam (d) None of these

- d. X band having a _____ frequency range. [01]

- (a) 4-8 GHz (b) 8-12 GHz
(c) 12-18 GHz (d) 8-13 GHz

- e. Define following terms. [03]

- 1) Minimum Detectable signal 2) Navigation term 3) unambiguous range

Q – 2.a Prove that the amplitude of the direction finding loop antenna is given by $\frac{2\pi EAN}{\lambda} (\cos\phi)$ [04]**Q – 2.b Answer any two. [10]**

- (i) Derive Basic Radar equation.
- (ii) What types of displays will be used in Radar? Explain any two in Detail.
- (iii) Consider Radar with pulse repetition frequency 1000 Hz. (a) Find the time duration

between two pulses. (b) Suppose an echo from a distant object is received 20 μ sec after a pulse is transmitted, what is the distance of the object from the radar? (c) Is there a second -time-around echo from this object?

Q - 3 Answer any two. [14]

- An MTI radar is operating with 0.1GHz with a PRF of 200 Hz. Calculate the lower three blind speeds.
- Explain filter characteristics of Delay line canceller.
- Two MTI radars are found to have a blind speed are given by $V_{b2}=20\text{m/s}$ of first MTI radar and second Radar at $V_{b3}=30\text{ m/s}$ of second MTI radar if they operate on wavelength of 30 cm find the ratio of PRF of the Radar.

SECTION – II

Q - 4 Answer the question below. [07]

- LORAN stands for_____. [01]
 - Long range navigational aid
 - Latest range navigational aid
 - Least range navigational aid
 - Long rest navigational aid
- LORAN-C radar operates in _____ frequency Band. [01]
 - 90-120KHz
 - 80-110KHz
 - 90-110KHz
 - 90-130KHz
- OMEGA system is _____system. [01]
 - Analog
 - Digital
 - Analog and Digital
 - Hybrid
- Draw Polar Diagram of Four Course Radio Range. [02]
- Justify Statement “TACAN overcomes the limitations of VOR” [02]

Q – 5.a Draw neat block diagram of FMCW radar and enlist its distinguished features. [05]

OR

Q – 5.a Draw and explain block diagram of FMCW altimeter. [05]

Q – 5.b What do you mean by Beat frequency explain in detail. [04]

Q – 5.c What is Doppler effect? Draw the block diagram of Pulse Doppler Radar. Explain function of each block in detail. [05]

OR

Q – 5.c How detection of multiple targets moving with different velocities has done by Doppler Radar? [05]

Q – 6.a Explain in brief the Distance Measuring Equipment. [05]

Q – 6.b Write a short note on Synthetic aperture Radar. **[05]**

Q – 6.c Derive the Doppler Radar Range Equation. **[04]**

OR

Q – 6.a Write a short note on Navy Radar. **[05]**

Q – 6.b Derive an expression for resolution element along the swath width of side looking Airborne Radar. **[05]**

Q – 6.c Explain in brief the Coherent Pulsed Doppler Radar system. **[04]**
